

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 07

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from:</i> • Discharge of tank-washing water from tankers • Oil rig / drilling accidents / leakage during drilling • Pipeline leaks • Leakage from storage tanks / refinery spills • Waste lubricating oil from vehicles / industry washed into drains • Other ship accidents (fuel oil)	3
01.2	• Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia • Feathers lose buoyancy / waterproofing so heavily oiled birds drown • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young	3
01.3	<i>Any 2 from:</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen • Toxic components (benzene, xylene) dissolve and poison plankton near the surface • Oil blocks light so algae / phytoplankton cannot photosynthesise • Oil masks odours / narcotises so animals cannot find mates or food by smell • Shore organisms (molluscs, crustaceans) are covered and asphyxiated	2
01.4	• Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	

Question	Marking guidance	Marks
02.1	- Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve - Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed - Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle	3
02.2	- Auger collects a core of known cross-sectional area and depth - So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume)	2
02.3	- Minerals such as calcium carbonate break down releasing carbon dioxide - So mass is lost that is not organic matter	2
02.4	- A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter - Standardise bulb brightness / colour, distance and container - A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
03.1	• $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$	1
03.2	- Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ - Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter	2
03.3	<i>Any 2 from:</i> - Low / zero tillage - Adding manure, compost or mulch - Growing cover crops / leaving crop residues - Converting to long-term grass or tree crops in rotation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
04.1	- Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead - The warm surface heats the air above; warm winds (eg south-westerlies) carry heat energy to higher latitudes - Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe - Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes; also transports water to areas that would be dry	4
04.2	- The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe - Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land	2

Question	Marking guidance	Marks
04.3	- Atmospheric pressure controls how easily water molecules evaporate / escape from the surface - At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour	2
04.4	<i>Any 2 from:</i> - Nitrogen - Oxygen - Carbon dioxide - Argon / neon / krypton / xenon	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	
05.1	$(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%)	1
05.2	- A fixed fee gives no incentive to save water because the bill is the same however much is used - Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances)	2
05.3	<i>Any 3 from:</i> - Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers - Greywater re-use: bath / shower water used to flush toilets or water gardens - Rainwater collection / water butts for garden use - Xeriscaping: planting drought-adapted species that need no watering - Repairing leaks / dripping taps	3
05.4	- Old infrastructure / pipes in poor condition leak - Repair and good maintenance / pipe replacement / leak detection reduce wastage	2
05.5	- Water is delivered directly to the plant roots / soil around each plant - Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
06.1	$(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$	1
06.2	- Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation - The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines	2
06.3	- Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) - Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) - Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge	3

Question	Marking guidance	Marks
06.4	<i>Any 4 from:</i> • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	• In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) • But not proportional: B has a high GDP but much lower energy use than A or E; C uses relatively high energy for its GDP	2
07.2	• Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes • Manufacture and transport of the extra products bought uses energy (indirect use); larger houses use more energy to build • Affluent countries can afford energy for non-essential uses; cheap local energy (eg petrol in the USA) encourages wasteful use	3
07.3	<i>Any 2 from:</i> • Higher energy prices / taxes encouraging conservation • Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness • Milder climate needing less heating or air conditioning • Service-based economy: heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted • Smaller / more efficient vehicles, better public transport, better insulated buildings	2
07.4	• Decline of heavy industries such as steel and aluminium smelting in the UK • The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country; also efficiency improvements	2
07.5	• Water: pumping, treatment / sterilisation, sewage treatment; Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	
08.1	• Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) • These react with oxygen in the air spaces between the waste particles (oxidation) • Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid	3

Question	Marking guidance	Marks
08.2	- The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons - Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) - Only a few tolerant taxa survive; the sensitive mayfly larvae (indicator of clean water) are eliminated; dissolved oxygen is not the limiting factor here	3
08.3	- Dilution by tributaries / more water - Neutralisation by alkaline rocks (limestone) in the stream bed; zinc precipitates / adsorbs onto sediments as pH rises	2
08.4	- Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid - Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	
09.1	- High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) - Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults - Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg)	6
09.2	- Radioactive decay of the concentrated fission products generates heat which could damage containers - Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years	2
09.3	- Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) - Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
10.2	• The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25